Electronic Standby Instrument System (ESIS) (Mod 6/2049 and Mod 6/2170) – General data

1. General

The Electronic Standby Instrument System (ESIS) gives displays for attitude, heading, altitude and airspeed in case of primary display or systems failures. The ESIS is powered by its own dedicated PS–850B backup battery, and is entirely independent of the Apex system. Unless noted otherwise, the descriptions and procedures in this section are common to both the GH-3100 and the GH-3900.2 ESIS units.

2. Description

A. GH–3100 or GH–3900.2 (Mod 6/2170) Electronic Standby Instrument System

The ESIS is a 3–ATI size instrument with no moving parts. It utilizes flat panel, Active Matrix Liquid Crystal Display (AMLCD) technology. The ESIS contains a 3 axis inertial measurement sensor cluster that is used to compute attitude information. Airspeed and altitude information is displayed on either side of the attitude display in a digital vertical scale. Magnetic heading is displayed at the bottom. Airspeed and altitude information is computed using an integrated air data computer module that is supplied with pitot and static pressure from the right hand pitot tube and the upper set of static vents. In the event of obstruction of the upper static vents, the ESIS STATIC SELECTOR may be moved to the EMERGENCY position. When this is done, static pressure from the lower static vents is supplied to the ESIS. The pitot pressure input to the ESIS is not switchable. The design of the display has been customized to replicate as closely as possible the display of information on the Apex PFD.

The ESIS is controlled by the ESIS POWER switch on the fuel panel. The switch has an ARM, OFF, and TEST position. The TEST position is used to do a battery capacity test, refer to Para 2.D for details. When the ESIS POWER switch is in the ARM position, the aircraft supplies necessary power to charge the battery pack and operate the ESIS. In the event of an aircraft power failure, the PS–850B automatically takes over the duties of supplying the main power input and lighting input to the ESIS and ensures an uninterrupted flow of power. Whenever power is applied to the ESIS, a complete internal self test sequence of the ESIS unit is carried out. Any faults found will be displayed on the ESIS screen.

28 Volt DC power is provided from the ESIS bus through a 3 amp ESIS INST (P5) circuit breaker on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

B. MAG–3000 Magnetometer

The magnetometer uses a three-axis magnetic sensor to sense magnetic fields. The field data is transmitted via an RS–422 bus to the ESIS. The data is used to compute the magnetic heading of the aircraft. The magnetometer is located in the outboard end of the left wing.

28 Volt DC power is provided from the ESIS bus through a 1 amp ESIS MAG (Q5) circuit breaker on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

C. Configuration Module

The configuration module contains sufficient non-volatile memory to retain all the programming codes that describe the specific hardware and software configuration of each installation, such as display format configuration. When the ESIS is removed from the aircraft, the configuration module remains in the aircraft. This eliminates the need to reprogram a new ESIS if the instrument is replaced.

The GH–3100 (Pre Mod 6/2170) uses a DCM–3100 Detachable Configuration Module (DCM), which attaches to the ESIS as the P2 connector. The DCM–3100 is tethered to the aircraft harness, to ensure it remains in the aircraft when the ESIS is removed.

The GH–3900.2 (Post Mod 6/2170) uses a Data Configuration Module (DCM), which is tied into the wiring harness and is wired directly into the P1 connector.

D. PS–850B Emergency Power Supply

The PS–850B Emergency Power Supply is a self-contained power supply mounted in the rear avionics compartment. It is used to power the ESIS bus in the event of total aircraft electrical power failure. When fully charged, the backup battery has sufficient capacity to operate the ESIS for approximately 6 hours.

The emergency power supply has a test feature which tests battery capacity. Using either the push button on the front panel of the power supply, or the ESIS POWER switch, TEST position places a 7 ohm load (battery heater) across the output of the battery. If the voltage under load is acceptable (50% or greater than total capacity) the LED will light and the battery will provide a ground to the MAU, which triggers the ESIS Batt Test Pass CAS message.

28 Volt DC power is provided from the left bus through a 15 amp ESIS BATTERY (N5) circuit breaker on the avionics circuit breaker panel (refer to 24-60-00, DC Electrical Load Distribution – General Data).

28 Volt DC power is provided to the ESIS bus through a 5 amp ESIS POWER circuit breaker located near the emergency power supply.

3. Adjustment/Test

A. Electronic Standby Instrument Test

CAUTION **DO NOT LEAVE THE ESIS POWER SWITCH IN THE ARM POSITION WHEN AIRCRAFT ELECTRICAL POWER IS NOT ON. BATTERY DAMAGE CAN OCCUR IF THE BATTERY DISCHARGES FULLY.**

1) ESIS Operational Test

Note No voltage shall be present on the 28 Volt DC input during test.

a) Ensure NO electrical power is applied to the aircraft.

b) Select ESIS POWER switch to the ARM position.

c) Verify the ESIS powers up, receiving electrical power ONLY from the emergency battery.

d) Verify the Built-in Test (BIT) and unit alignment complete (up to 3 minutes).

Note If any of the following are out of tolerance, refer to Para 5.A.

e) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

f) Allow avionics to fully initialize including ADAHRS alignment.

g) Compare the ESIS to the PFD's and verify the following:

1 Attitude will be ± 1.5°

2 Heading will be ± 6.0°

3 Airspeed will be at minimum setting (30 KIAS for GH–3100 and blank airspeed field for GH–3900.2)

4 Altitude will be ± 20 feet (ensure proper baro setting)

h) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

i) Verify the ESIS continues normal operation.

j) Select ESIS POWER switch to the OFF position.

B. PS–850B Battery Check

1) Periodic Battery Check

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical systems to initialize.

c) Select ESIS POWER switch to the momentary TEST position and hold.

d) After approximately 5 seconds, the ESIS Batt Test Pass CAS message should appear.

e) If no CAS message appears, do the Battery Capacity Test as per Para 3.B.2).

f) Select ESIS POWER switch to the OFF position.

g) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Battery Capacity Test

a) Ensure the aircraft electrical system is OFF, and ESIS POWER switch is in the OFF position.

b) Gain access to the Emergency Power Supply, which is accomplished through the aft baggage compartment into the rear avionics compartment.

c) Depress and hold the TEST switch on the front panel of the unit for 5 seconds.

1 If the LED on the front panel is illuminated for the entire duration of the tests, the battery capacity charge is greater than 50%.

2 If the LED on the front of the panel extinguishes before the switch is released, or does not illuminate at all, the battery capacity charge is lower than 50%.

d) Battery requires recharging as per Para 3.B.4).

Note 1 If the battery capacity charge is greater than 50% – battery requires up to 8 hours of recharging.

2 If the battery capacity charge is lower than 50% – battery requires up to 16 hours of recharging.

e) Do a discharge test, refer to Para 3.B.3).

3) Discharge Test

Discharge test is done to determine if the ESIS Emergency Battery requires servicing. Battery to be tested must be at full capacity.

Equipment required:

- Digital voltmeter Fluke Model 8050A or equivalent
- 7 Ohm Load Resistor 7Ω + 1% 150 watt (minimum) ceramic wire wound WireMasters Inc.
- WM910053 or equivalent
- Stopwatch

a) Place the digital voltmeter across pins J1–11 (+) and J1–7 (–) on the ESIS Emergency Battery.

b) Place the 7Ω load resistor across pins J1–11 (+) and J1–7 (–) on the ESIS Emergency Battery. Start Stopwatch.

c) Time the discharge period. Output voltage observed on the digital voltmeter should be greater than 20 VDC at 48 minutes.

d) If output is lower than 20 VDC, the battery requires servicing at an approved avionics facility. If output is greater than 20 VDC, continue to Para 3.B.3).e).

e) Remove the digital voltmeter and load resistor from the ESIS Emergency Battery pins and recharge battery (refer to Para 3.B.4)) within 2 hours of completing discharge test.

4) Recharging the Battery

WARNING **DO NOT CHARGE A BATTERY ON THE AIRCRAFT. THERE MAY BE A SERIOUS RISK OF INJURY TO PERSONNEL AND/OR DAMAGE TO THE AIRCRAFT OR AIRCRAFT SYSTEMS.**

Refer to the battery manufacturer's information for detail procedures, alternative methods of charging and procedures for storage.

Equipment required:

- DC Power Supply Hewlett Packard Model 6291 or equivalent
- Digital Voltmeter Fluke Model 8050A or equivalent
- Local Manufactured wire harness comprised of:

  • Mating Connector DPXB–13–33S–001 or equivalent
  • Isolation Diode 15 amp, 50 Volt PIV WireMasters Inc. WM1N3208 or equivalent
  • Wire 16 AWG

a) Remove ESIS battery from aircraft, refer to Para 4.D.1).

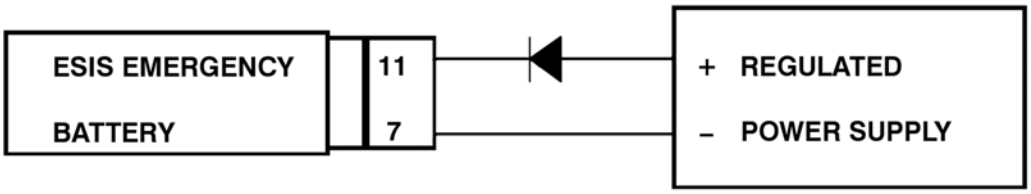
b) Set the voltage 29.0 ± 0.5 VDC on the regulated power supply. Ensure current is limited to 10 amp max, as shown below.

c) Connect power supply to battery using the local manufactured wiring harness.

d) Start regulated power supply, charge battery up to 16 hours.

Note
- Batteries are 80% capacity after 1 hour charging
- Batteries are at full capacity after 16 hours charging
- Charging for 23 hours is recommended for optimum cell balancing and maximum capacity

e) Stop regulated power supply and disconnect wiring harness from battery.



CAUTION **USE PIN J1–11 INSTEAD OF PIN J1–10 DURING CHARGING PROCEDURES TO AVOID ACTIVATING THE INTERNAL HEATING BLANKET.**

f) Reinstall ESIS battery in aircraft, refer to Para 4.D.2).

C. Electronic Standby Instrument System Functional Test

1) ESIS Function Test

a) Select ESIS POWER switch to the ARM position.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Allow the Built-In Test (BIT) to finish, and the unit alignment to complete (approximately 3 minutes).

Note If any of the following are out of tolerance, refer to Para 5.A.

d) Compare the ESIS to the PFD's and verify the following:

1 Attitude will be ± 1.5°

2 Heading will be ± 6.0°

3 Airspeed will be at minimum setting (30 KIAS for GH–3100 and blank airspeed field for GH–3900.2)

4 Altitude will be ± 20 feet (ensure proper baro setting)

e) Set the barometric correction to “STD” (29.92 in hg) by pressing the indicator adjustment knob.

f) Connect the air data test set the aircraft pitot and static ports (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4. Adjustment/Test).

g) Set the air data test set to achieve a 1000 – 3000 fpm climb. Ensure the altitude tape moves smoothly and displays the correct altitude.

h) Stop the test set at 5,000, 10,000, and 25,000 ft. Indicated altitude should match the following:

1 5,000 ft ± 25 ft

2 10,000 ft ± 30 ft

3 25,000 ft ± 50 ft

i) Set the air data test set to achieve an increasing airspeed. Ensure the airspeed tape moves smoothly and displays the correct airspeed. Indicated airspeed should match the following:

1 60 kts ± 5.0 kts

2 100 kts ± 2.0 kts

3 150 kts ± 2.0 kts

j) Adjust airspeed to greater than aircraft V_{MO}, a red bar will be displayed from V_{MO} to the high-end of the airspeed tape. Ensure the airspeed readout digits change to red when the airspeed is increased greater than V_{MO}.

k) Set the air data test set to achieve a 1000 – 3000 fpm descent. Ensure the altitude tape moves smoothly and displays the correct altitude.

l) Set the air data test set to achieve a decreasing airspeed. Ensure the airspeed tape moves smoothly and displays the correct airspeed.

m) Vent the pitot and static pressure to ambient conditions. Confirm the indicator displays field altitude and minimum airspeed setting (30 KIAS for GH–3100 and blank airspeed field for GH–3900.2).

n) Disconnect the air data test set from aircraft pitot and static ports (refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data).

o) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

p) Set the ESIS POWER switch to OFF.

2) ESIS Display Cleaning

The ESIS bezel, softkey and display glass can be cleaned with a damp lint-free, static-free cloth. If required, clean with a soft cloth moistened with a mild solution of soap and water. Care should be taken to avoid scratching the surface of the display or getting water inside the softkey.

D. Calibration

Note Deleted

1) Attitude Calibration

a) ESIS GH–3100 (Pre Mod 6/2170)

Note The attitude calibration does not change the attitude display. These procedures allow compensation of attitude differences between the indicator and magnetometer for improved calibration.

Refer to Section 3.2.2.2.1 of the GH–3100 Electronic Standby Instrument System Installation and Operation Manual, part number TP–559, at the latest applicable revision.

b) ESIS GH–3900.2 Panel Angle (Post Mod 6/2170)

Note The nominal panel installation angle is set within the configuration file, and normally should not need adjustment. The following procedure should only be used if adjustment is deemed necessary.

1 Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2 To access the ESIS maintenance menu, press and hold the M key and BARO knob; select ESIS POWER switch to ARM.

3 Wait several minutes for the aircraft electrical and avionics systems to initialize.

4 Turn on the maintenance laptop computer and boot up to the desktop.

5 Connect the maintenance laptop computer to the ADAHRS using an RS–232 maintenance cable from an available computer COM port, to the ADAHRS 1B maintenance port on the aircraft maintenance panel located in the forward avionics bay.

6 Start the KSG 7200 Maintenance Program on the maintenance laptop computer.

7 Select the appropriate COM port and click ‘Start Maintenance’.

8 Click ‘Attitude’ to bring up a window with the digital pitch and roll values for ADAHRS channel B. Wait for pitch and roll values to stabilize.

9 Within the ESIS maintenance menu, ensure ANGLES is highlighted, press the BARO knob to enter the ANGLES menu.

10 When the ANGLES menu is first entered, the MOUNT ANGLE field displays the panel mounting angles stored in the configuration module.

11 Rotate the BARO knob to select pitch or roll AIRCRAFT ATTITUDE data fields.

12 Press the BARO knob to select the field, rotate the BARO knob to adjust the value in each field, then press BARO to enter the selected value.

13 Adjust the AIRCRAFT ATTITUDE pitch angle until the CALC AC ANGLE pitch value matches the ADAHRS pitch value displayed on the maintenance computer.

14 Adjust the AIRCRAFT ATTITUDE roll angle until the CALC AC ANGLE roll value matches the ADAHRS roll values displayed on the maintenance computer.

15 Press the M button to exit the ANGLES menu. Press BARO to select SAVE CHANGES before returning to the SETUP menu.

16 On the maintenance laptop, click 'Close'.

17 Click 'Exit Maintenance'.

18 Close the maintenance program on the laptop maintenance computer.

19 Remove the maintenance laptop computer and RS-232 maintenance cables from the aircraft.

20 Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

21 Set the ESIS POWER switch to OFF.

c) ESIS GH–3900.2 Magnetometer Angle (Post Mod 6/2170)

1 Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2 To access the ESIS maintenance menu, press and hold the M key and BARO knob; select ESIS POWER switch to ARM.

3 Rotate the BARO knob to select MAG INSTALL, press the BARO knob to enter the MAG INSTALL menu.

4 From the MAG INSTALL menu, use the BARO knob to select MOUNTING.

5 Rotate the BARO knob to highlight desired parameter.

6 Push the knob to select the item, then rotate the knob clockwise to increment, and counterclockwise to decrement the value. Enter the magnetometer installation settings (refer to Table 1 — Magnetometer Installation Settings).

Table 1 Magnetometer Installation Settings

Parameter	Setting
PITCH	1.9°
ROLL	2.8°
YAW	0.0°

7 Press the M button to exit the MAG MOUNTING screen and return to the MAG INSTALL menu.

8 From the MAG INSTALL menu, use the adjustment knob to select WING FLEX COMPENSATION.

9 Rotate the BARO knob to highlight desired parameter.

10 Push the knob to select the item, then rotate the knob clockwise to increment, and counter clockwise to decrement the value. Enter the wing flex compensation settings (refer to Table 2 — Wing Flex Compensation Settings.

Table 2 Wing Flex Compensation Settings

Parameter	Setting
NO COMPENSATION	20 kts
FULL COMPENSATION	40 kts
PITCH COMPENSATION	0.0°
ROLL COMPENSATION	1.3°

11 Press the M button to exit the WING FLEX COMPENSATION screen and return to the MAG INSTALL menu. Press the M button (EXIT) to enter the MAG INSTALL SAVE menu.

12 Press the SAVE button (SAVE) to save changes or press the M button (EXIT) to exit without saving before returning to the SETUP MENU.

13 Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

14 Set the ESIS POWER switch to OFF.

2) Heading Calibration

a) ESIS GH–3100 (Pre Mod 6/2170)

Refer to Section 3.2.2.2.2 of the GH–3100 Electronic Standby Instrument System Installation and Operation Manual, part number TP–559, at the latest applicable revision.

b) ESIS GH–3900.2 (Post Mod 6/2170)

It is recommended that the entire procedure be reviewed prior to beginning MAG swing.

Note
- All installation angles (ANGLES menu and MAG MOUNTING menu) must be entered and saved, and the indicator must be re-booted prior to starting the MAG swing procedure.
- The MAG swing procedure must be done in an open area that allows the aircraft engines to be running, and can be safely swung.
- The MAG swing procedure must be done in a similar ambient magnetic field on the ground, as exists when the aircraft is in-flight (e.g. aircraft engine(s) should be running and all avionics equipment should be powered).
- Avoid being in the proximity of objects that may alter the magnetic field around the Magnetometer (e.g. rebar in concrete, metal building, etc).

1 Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2 Start both engines (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

3 Position the aircraft in an area (preferably a compass rose) free of distortions (no rebar), in the earth's magnetic field where it can swing 360°.

4 To access maintenance menu, press and hold the M key and adjustment knob; select ESIS POWER switch to ARM.

5 Rotate the adjustment knob to select MAG SWING; press the adjustment knob to enter the MAG SWING menu. A PRESS KNOB TO START message is shown at the top of the screen if valid magnetometer data is being received by the indicator.

6 With the aircraft in position (on any heading) and stationary, press the adjustment knob to begin alignment.

7 A WAIT message is shown at the top of the screen and an ALIGNING message is shown below it. The bar to the right of the ALIGNING message represents the alignment progress.

8 When the WAIT message is replaced with 360° message, begin swinging the aircraft. The ALIGNING message is replaced by SWINGING. The bar to the right of the SWINGING message represents the progress of the swing procedure.

9 Swing the aircraft in a complete 360° clockwise turn, until VERIFYING message is shown. During the swing procedure, monitor the Turn Rate bar. The pointer (white triangle) must not enter the red area of the bar.

Note
- If the pace of the swing is too fast (greater than 9° per second), a FAILED message is shown at the top of the screen resulting in a brief OVERRATE message, followed by the removal of the SWINGING message and progress bar.
- If the pace of the swing is too slow (takes longer than 300 seconds), a FAILED message is shown at the top of the screen resulting in the removal of the SWINGING message and progress bar.
- If a problem with magnetometer data is detected during the swing procedure, a FAILED message is shown at the top of the screen.
- If FAILED message is witnessed, restart the procedure from Para 3.D.2).b).5.

10 When the MAG swing procedure is complete, the progress bar is removed and a VERIFYING message is shown at the top of the screen.

11 If the FIELD VAR is less than or equal to 5mG, the message SWING OK is shown at the top of the screen. If the FIELD VAR is greater than 5mG, the message FAILED is shown at the top of the screen. If FAILED message is witnessed, restart the procedure fromPara 3.D.2).b).5

12 Press the M button to exit the MAG SWING menu. Select SAVE CHANGES when returning to the SETUP MENU.

13 Cycle power to the GH–3900.2 to apply swing data to unit.

14 Do a calibration check, refer to Para 3.D.3).

3) Calibration Check

a) Select ESIS POWER switch to the ARM position.

b) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

c) Start both engines (refer to 71-00-00, Power Plant (Mod 6/2012) – General Data).

d) Allow ESIS to complete self-test/alignment process.

e) Position the aircraft in an area (preferably a compass rose) free of distortions (no rebar), in the earth's magnetic field where it can swing 360°.

Note To speed up the following steps, when aligned at the desired heading, instead of waiting 3 minutes for the heading to stabilize, initiate a manual alignment by selecting "Alignment >Initiate Alignment" from the ESIS menu.

f) Align the aircraft to a cardinal heading (e.g. North, East, South, or West). Allow the heading to stabilize by waiting a minimum of 3 minutes. Ensure heading is within ±4.0° of the known magnetic heading.

g) Align the aircraft to a heading 45° apart from first position. Allow the heading to stabilize by waiting a minimum of 3 minutes. Ensure heading is within ±4.0° of the known magnetic heading.

h) Align aircraft to remaining headings at 45° increments. Allow the heading to stabilize by waiting a minimum of 3 minutes. Ensure heading is within ±4.0° of the known magnetic heading.

E. Gear Configuration Change

1) ESIS GH–3100 (Pre Mod 6/2170)

The GH–3100 incorporates a Detachable Configuration Module (DCM) to change between various gear configurations.

To switch configurations, do the following:

a) Remove existing DCM, refer to Para 4.C.1).

b) Install new DCM, refer to Para 4.C.2).

c) Do a configuration verification, refer to Para 3.F.

2) ESIS GH–3900.2 (Post Mod 6/2170)

The GH–3900.2 incorporates a Data Configuration Module (DCM), tied into the ESIS wiring harness, to store gear configuration and a maintenance connector to load various gear configurations.

To switch configurations, do the following:

a) Apply external power to aircraft (refer to 12-00-00, Servicing – General Data).

b) To access MAINTENANCE menu, press and hold the M key and adjustment knob; select ESIS POWER switch to ARM.

c) Rotate the adjustment knob to select DATA LOAD; press the adjustment knob to enter the DATA LOAD menu.

d) Press the adjustment knob to Activate Data Load and place the unit in Date Load Mode.

e) Connect the maintenance computer to the ESIS connector on the maintenance panel using a serial cable, or RS–232/USB converter.

f) On the maintenance computer, start the GH–3900.2 Field Loader software (Part Number 576–6158–02).

g) Select the appropriate serial (COM) port, press the Connect button to establish communication between the GH–3900.2 and the maintenance computer.

h) Press the Browse button and locate desired configuration file (.bin file extension) that matches the aircraft gear configuration. Select the file and press the Update button.

i) During the transfer of data the indicator displays “Data Load: TRANSFER IN PROGRESS”.

j) If the transfer of data is successful, the indicator shows the message “Data Load: TRANSFER COMPLETE, READY”.

Note If an error message is displayed, refer to Section 3.3.1.1 of GH–3900.2 Installation Manual, part number 0040–34401–01, at the latest applicable revision.

k) Press the Disconnect button on the GH–3900.2 Field Loader software and remove RS–232 cable.

l) Select ESIS POWER switch to OFF. Remove external power from aircraft.

m) Do a configuration verification, refer to Para 3.F.

F. Configuration Verification

1) Select ESIS POWER switch to the ARM position.

2) When ESIS splash screen appears, note the aircraft model name, DHC-6-400 XXXXXx. (XXXXXX = LAND, SKIS, or FLOATS).

3) Confirm LAND, SKIS, or FLOATS matches the current gear configuration.

G. Altimeter System Test

1) Select ESIS POWER switch to the ARM position.

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Wait several minutes for the aircraft electrical system to initialize and the altitude flags to clear.

4) Leak test the Pitot-Static System. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test).

5) Ensure that the (ESIS) STATIC SOURCE valve is in the NORMAL position.

6) Connect the air data test set to the RH pitot head and upper static port on the aircraft. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data, Para 4., Adjustment/Test).

7) Complete the Altitude Calibration Table (refer toTable 3 — Altitude Calibration Table).

Table 3 Altitude Calibration Table

TEST SET ALTITUDE (FEET)	TOLERANCE (± FEET)	RECORD ERROR	REMARKS	INITIALS
		ESIS		
– 1,000	20			
0	20			
500	20			
1,000	20			
1,500	25			
2,000	25			
3,000	25			
4,000	25			
5,000	25			
6,000	25			
8,000	30			
10,000	30			
12,000	35			
14,000	40			
16,000	40			
18,000	45			

Table 3 Altitude Calibration Table (continued)

TEST SET ALTITUDE (FEET)	TOLERANCE (± FEET)	RECORD ERROR	REMARKS	INITIALS
		ESIS		
20,000	50			
22,000	50			
25,000	50			

4. Removal/Installation

WARNING INSTALLATION OF THIS COMPONENT CONTAINS PROTECTION FEATURES THAT ARE CRITICAL TO PREVENTING DEGRADATION FROM THE EFFECTS OF HIGH INTENSITY RADIATED FIELDS (HIRF) AND INDIRECT EFFECTS OF LIGHTNING (IEL). ENSURE INTEGRITY OF THE HIRF/IEL PROTECTION FEATURES ARE MAINTAINED IN ACCORDANCE WITH THE FOLLOWING:

CHASSIS GROUNDING – PANEL/PEDESTAL MOUNT UNITS CANNOT BE RELIABLY GROUNDED TO THE AIRFRAME VIA THE PANEL MOUNTING SCREWS OR DZUS FASTENERS. GROUNDING OF THESE UNITS IS ACCOMPLISHED BY USING THE SHIELD GROUNDING BRAID OR JUMPER. VERIFY PROPER ELECTRICAL BONDING AS DETAILED IN 20-75-00, ELECTRICAL BONDING AND GROUNDING OF AIRCRAFT STRUCTURES – GENERAL DATA.

SHIELD GROUNDS – VERIFY SHIELD GROUNDING AT CONNECTOR BACKSHELLS AS PER REQUIREMENTS IN PSM 1-64-2W.

BACKSHELLS OR STRAIN RELIEFS – VERIFY PROPER ELECTRICAL BONDING BETWEEN THE BACKSHELL/STRAIN RELIEF AND THE CONNECTOR AS PER REQUIREMENTS IN PSM 1-64-2W.

Note The Electronic Standby Instrument System has some additional shielding requirements; refer to the notes on the applicable wiring diagram.

A. Electronic Standby Instrument System (ESIS)

1) Removal of ESIS

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the ESIS INST (P5) circuit breaker and fit safety clip.

c) Loosen the (2) large clamp screws on the top right and bottom left corners of the indicator.

d) Carefully pull the ESIS out of the instrument panel to give access to the electrical connector and the pitot and static fittings.

e) Disconnect electrical connector (P287) and configuration module (P288). Ensure configuration module chain is attached to electrical harness.

f) Disconnect pitot and static fittings on rear of instrument.

g) Remove instrument from aircraft.

Note Stow connectors and fittings as required.

2) Installation of ESIS

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the ESIS INST (P5) circuit breaker is opened and fitted with a safety clip.

c) Hold the ESIS near its mount in the instrument panel.

d) Connect the pitot and static fittings to the ESIS.

e) Connect the electrical connector (P287) and configuration module (P288) to the ESIS.

f) Slide the ESIS fully into its mount.

g) While pushing on the ESIS towards the instrument panel, tighten the (2) large clamp screws.

h) Remove the safety clip and close the ESIS INST (P5) circuit breaker.

i) Do pitot and static leak tests. (Refer to 34-11-00, Pitot-Static System (Mod 6/2033) – General Data).

j) Do an ESIS Function Test, refer to Para 3.C.1).

B. Magnetometer

CAUTION **DO NOT EXPOSE THE MAGNETOMETER TO MAGNETIC FIELDS GREATER THAN 4 GAUSS DURING PACKAGING, STORAGE, OR TRANSPORT IN ORDER TO PREVENT MAGNETIZATION OF THE MAGNETOMETER.**

1) Removal of Magnetometer

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the ESIS MAG (Q5) circuit breaker and fit safety clip.

c) Gain access to the magnetometer by removing the (4) non-ferrous mounting screws from the access panel in the underside of the left wing. (Refer to 12-00-00, Servicing – General Data).

d) Disconnect electrical connector (P286).

Note Stow connectors as required.

e) Remove the (3) non-ferrous magnetometer mounting screws.

f) Lower Magnetometer from shelf while feeding the pigtail through the opening.

2) Installation of Magnetometer

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the ESIS INST (P5) circuit breaker is opened and fitted with a safety clip.

c) Feed connector pigtail through opening in mounting shelf and connect electrical connector.

d) Install the (3) non-ferrous magnetometer mounting screws.

e) Install access panel with (4) non-ferrous mounting screws.

f) Remove the safety clip and close ESIS MAG (Q5) circuit breaker.

g) Do an ESIS calibration check, refer to Para 3.D.3). If check fails, do an ESIS heading calibration, refer to Para 3.D.2).

C. Detachable Configuration Module (Pre Mod 6/2170)

1) Removal of Detachable Configuration Module

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the ESIS INST (P5) circuit breaker and fit safety clip.

c) Loosen the (2) large clamp screws on the top right and bottom left corners of the indicator.

d) Carefully pull the ESIS out of the instrument panel to give access to the configuration module (P288).

e) Disconnect configuration module (P288).

f) Detach configuration module chain from electrical harness.

2) Installation of Detachable Configuration Module

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the ESIS INST (P5) circuit breaker is opened and fitted with a safety clip.

c) Attach configuration module chain to electrical harness.

d) Connect configuration module (P288) to J2 of the ESIS.

e) Remove the safety clip and close ESIS INST (P5) circuit breaker.

f) Do an ESIS attitude calibration, refer to Para 3.D.1).

g) Do an ESIS heading calibration, refer to Para 3.D.2).

D. ESIS Emergency Battery (PS-850B)

1) Removal of ESIS Emergency Battery

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open the ESIS BATTERY (N5) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Gain access to the rear avionics shelves through the rear baggage compartment.

d) Loosen retaining screw clamp from the front and slide the unit out of the ATR Rack.

e) Remove unit from aircraft.

Note Stow connectors as required.

f) Visually inspect ATR Rack. If damaged, replace as per Para 4.E.1).

2) Installation of ESIS Emergency Battery

Note If ESIS emergency battery has been in storage or, the battery capacity is unknown, do the Battery Capacity Test detailed in Para 3.B.3) prior to installation into aircraft.

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure the ESIS BATTERY (N5) circuit breaker is opened and fitted with safety clip.

c) Slide the unit onto the ATR Rack. Tighten the retaining screw clamp at the front lip of the ESIS Emergency Battery unit.

d) Remove safety clip and close ESIS BATTERY (N5) circuit breaker on the avionics circuit breaker panel.

e) Do the Periodic Battery Check, refer to Para 3.B.1).

f) Do the ESIS Operational Test, refer to Para 3.A.1).

E. ESIS Emergency Battery ATR Rack

1) Removal of ESIS Emergency Battery ATR Rack

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the rear avionics shelves through the rear baggage compartment.

c) Remove ESIS Emergency Battery from ATR Rack, refer to Para 4.D.1).

d) Remove the (4) screws that secure the ATR Rack to the bottom of the avionics rack.

e) Remove the (4) mounting screws from the electrical connector. Remove connector from the ATR Rack.

f) Remove ATR Rack from aircraft.

2) Installation of ESIS Emergency Battery ATR Rack

a) Ensure the aircraft electrical system if OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Install (4) mounting screws to secure the electrical connector to the ATR Rack.

c) Align the ATR Rack to the mounting holes on the rear avionics rack.

d) Install the four screws that secure the ATR Rack to the avionics rack.

5. Troubleshooting

A. Electronic Standby Instrument System

1) Ensure ESIS BATTERY (N5), ESIS INST (P5), and ESIS MAG (Q5) circuit breakers are closed.

2) Visually inspect ESIS and magnetometer connectors for damaged wiring or contacts. Ensure electrical connector on ESIS and magnetometer are fully seated and locked.

3) Do a pitot/static leak test. Refer to 34-11-00, Pitot Static System (Mod 6/2033) – General Data.

4) Ensure STANDBY INSTRUMENT STATIC SOURCE lever is in the NORMAL position.

5) Ensure pitot and static fittings are secure.

6) As applicable to the ESIS type installed, refer to Section 4.3 of the GH–3100 Electronic Standby Instrument System Installation and Operation Manual, part number TP–559, or Section 4.3 of the GH–3900.2 Electronic Standby Instrument System Installation and Operation Manual, part number 0040–34401–01, at the latest applicable revision.

7) If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in Wiring Diagrams Manual PSM 1-64-2W.

8) If repair or service is required, send to appropriately rated avionics repair facility.

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